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Specification For HINK 10.2''EPD

Model NO.: HINK-E102A03

Product VER:A0

Customer Approval

Customer	
Approval By	
Date Of Approval	

It will be agreed by the receiver,if not sign back the Specification within 15days.

Prepared By	Checked By	Approval By
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Version	Content	Date	Producer
A0	New release	2019/3/8	Wang lin



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1. General Description

HINK-E102A03 is an Active Matrix Electrophoretic Display (AMEPD), with interface and a reference system design. The 10.2" active area contains 960×640 pixels, and has 1-bit B/W full display capabilities. An integrated circuit contains gate buffer, source buffer, interface, timing control logic, oscillator, DC-DC. SRAM.LUT, VCOM and border are supplied with each panel.

2. Features

- 960×640 pixels display
- High contrast
- High reflectance
- Ultra wide viewing angle
- Ultra low power consumption
- Pure reflective mode
- Bi-stable display
- Commercial temperature range
- Landscape, portrait modes
- Hard-coat antiglare display surface
- Ultra Low current deep sleep mode
- On chip display RAM
- Low voltage detect for supply voltage
- High voltage ready detect for driving voltage
- Internal temperature sensor
- 10-byte OTP space for module identification
- Waveform stored in On-chip OTP
- Serial peripheral interface available
- On-chip oscillator
- On-chip booster and regulator control for generating VCOM, Gate and Source driving voltage
- I2C signal master interface to read external temperature sensor/built-in temperature sensor

3. Application

Electronic Shelf Label System

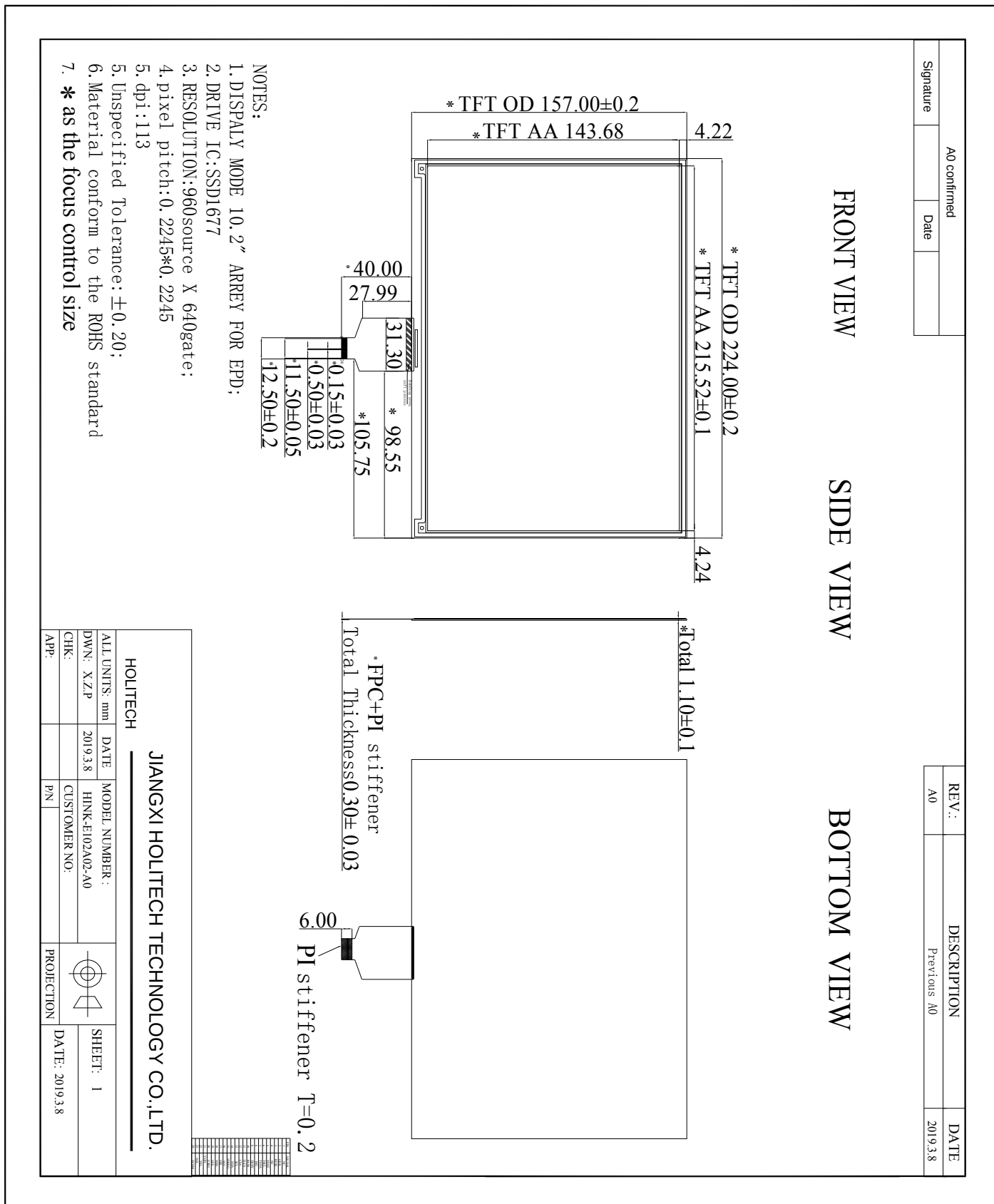
4. Mechanical Specifications

Parameter	Specifications	Unit	Remark
Screen Size	10.2	Inch	
Display Resolution	640(H)×960(V)	Pixel	Dpi:113
Active Area	215.52(H)×143.68(V)	mm	
Pixel Pitch	0.2245×0.2245	mm	
Pixel Configuration	Rectangle		
Outline Dimension	224(H)×157 (V) ×1.10(D)	mm	Without masking film
Weight	60±0.2	g	



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5. Mechanical Drawing of EPD module





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6. Input/Output Terminals

Pin #	Single	Description	Remark
1	NC	No connection and do not connect with other NC pins	Keep Open
2	GDR	N-Channel MOSFET Gate Drive Control	
3	RESE	Current Sense Input for the Control Loop	
4	NC	No connection and do not connect with other NC pins e	Keep Open
5	VSH2	Positive Source driving voltage	
6	TSCL	I2C Interface to digital temperature sensor Clock pin	
7	TSDA	I2C Interface to digital temperature sensor Date pin	
8	BS1	Bus selection pin	Note 6-5
9	BUSY	Busy state output pin	Note 6-4
10	RES #	Reset	Note 6-3
11	D/C #	Data /Command control pin	Note 6-2
12	CS #	Chip Select input pin	Note 6-1
13	SCL	serial clock pin (SPI)	
14	SDA	serial data pin (SPI)	
15	VDDIO	Power for interface logic pins	
16	VCI	Power Supply pin for the chip	
17	VSS	Ground	
18	VDD	Core logic power pin	
19	VPP	Power Supply for OTP Programming	
20	VSH1	Positive Source driving voltage	
21	VGH	Power Supply pin for Positive Gate driving voltage and VSH	
22	VSL	Negative Source driving voltage	
23	VGL	Power Supply pin for Negative Gate driving voltage, VCOM and VSL	
24	VCOM	VCOM driving voltage	



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Note 6-1: This pin (CS#) is the chip select input connecting to the MCU. The chip is enabled for MCU communication: only when CS# is pulled LOW.

Note 6-2: This pin (D/C#) is Data/Command control pin connecting to the MCU. When the pin is pulled HIGH, the data will be interpreted as data. When the pin is pulled LOW, the data will be interpreted as command.

Note 6-3: This pin (RES#) is reset signal input. The Reset is active low.

Note 6-4: This pin (BUSY) is Busy state output pin. When Busy is High the operation of chip should not be interrupted and any commands should not be issued to the module. The driver IC will put Busy pin High when the driver IC is working such as:

- Outputting display waveform; or
- Communicating with digital temperature sensor

Note 6-5: This pin (BS1) is for 3-line SPI or 4-line SPI selection. When it is “Low”, 4-line SPI is selected. When it is “High”, 3-line SPI (9 bits SPI) is selected.

7. MCU Interface

7.1 MCU interface selection

E102A03 can support 4-wire or 3-wire serial peripheral MCU interface, which is pin selectable by BS1 pin. The interface pin assignment for different MCU interfaces is shown in Table 7-1.

Note

- (1) L is connected to VSS
- (2) H is connected to VDDIO

Table 7-1: Interface pin assignment for different MCU interfaces

MCU Interface	Pin Name						
	BS1	RES#	CS#	D/C#	SCL	SDI	SDO
4-wire serial peripheral interface (SPI)	L	Required	Required	Required	SCL	SDI	SDO
3-wire serial peripheral interface (SPI) – 9 bits SPI	H	Required	Required	L	SCL	SDI	SDO

7.2 MCU Serial Peripheral Interface (4-wire SPI)

The 4-wire SPI consists of serial clock SCL, serial data input SDI, D/C# and CS#. The control pins status in 4-wire SPI in writing command/data is shown in Table 7-2 and the write procedure in 4-wire SPI is shown in Figure 7-1..

Table 7-2 : Control pins status of 4-wire SPI

Function	SCL pin	SDI pin	D/C# pin	CS# pin
Write command	↑	Command bit	L	L
Write data	↑	Data bit	H	L



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Note:

- (1) L is connected to VSS and H is connected to VDDIO
- (2) ↑ stands for rising edge of signal
- (3) SDI is shifted into an 8-bit shift register on every rising edge of SCL in the order of D7, D6, ... D0. The level of D/C# should be kept over the whole byte. The data byte in the shift register is written to the Graphic Display Data RAM (RAM)/Data Byte register or command Byte register according to D/C# pin.

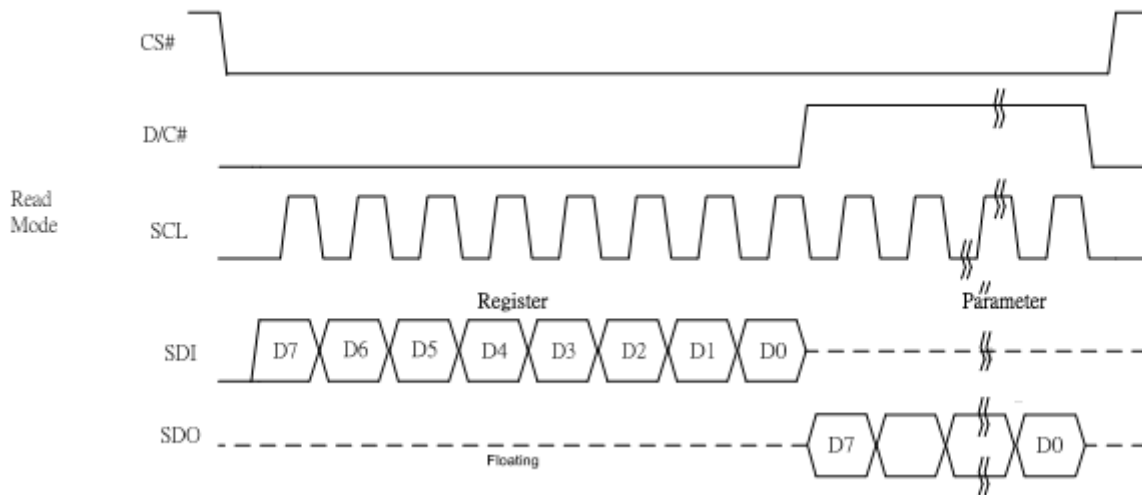


Figure 7-1: wire mode

In the read operation, after CS# is pulled low, the first byte sent is command byte, D/C# is pulled low. After command byte sent, the following byte(s) read are data byte(s), so D/C# bit is then pulled high. An 8-bit data will be shifted out on every clock falling edge. The serial data output SDO bit shifting sequence is D7, D6, to D0 bit. Figure 6-2 shows the read procedure in 4-wire SPI.

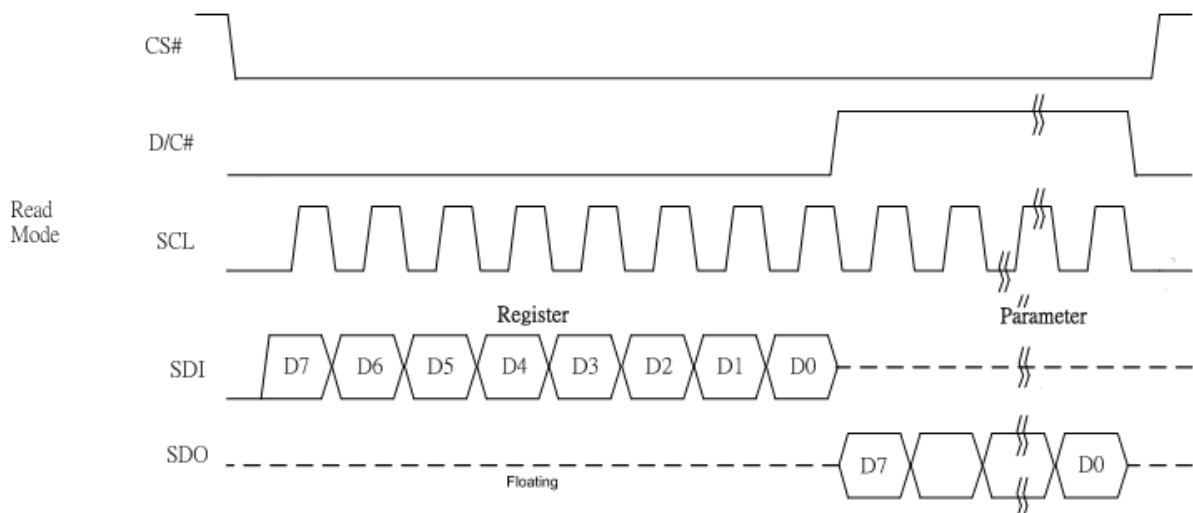


Figure 7-2: Read procedure in 4-wire SPI mode



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7.3 MCU Serial Peripheral Interface (3-wire SPI)

The 3-wire SPI consists of serial clock SCL, serial data input SDI, and CS#. The operation is similar to 4-wire SPI while D/C# pin is not used and it must be tied to LOW. The control pins status in 3-wire SPI is shown in Table 7-3.

In the write operation, a 9-bit data will be shifted into the shift register on every clock rising edge. The bit shifting sequence is D/C# bit, D7 bit, D6 bit to D0 bit. The first bit is D/C# bit which determines the following byte is command or data. When D/C# bit is 0, the following byte is command. When D/C# bit is 1, the following byte is data. Table 7-3 shows the write procedure in 3-wire SPI

Table 7-3 : Control pins status of 3-wire SPI

Function	SCL pin	SDI pin	D/C# pin	CS# pin
Write command	↑	Command bit	Tie LOW	L
Write data	↑	Data bit	Tie LOW	L

Note:

- (1) L is connected to V_{SS} and H is connected to V_{DDIO}
- (2) ↑ stands for rising edge of signal

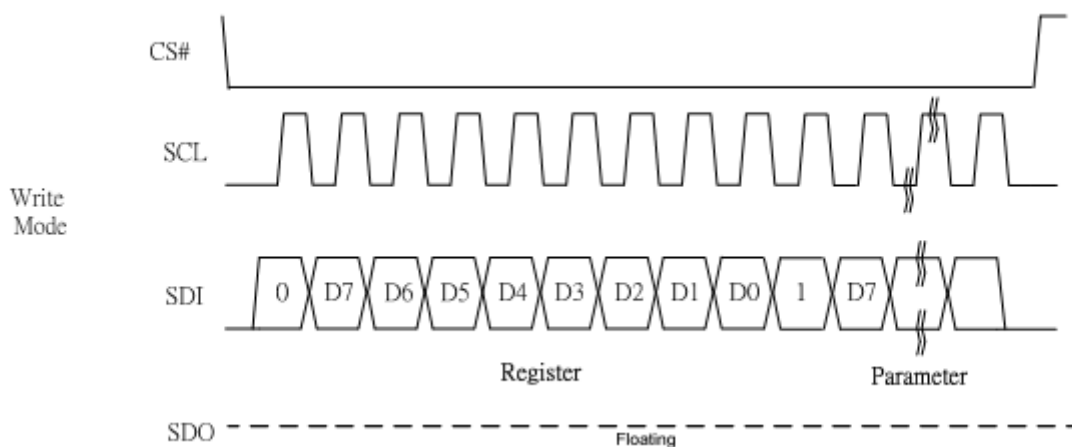


Figure 7-3: Write procedure in 3-wire SPI mode



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In the read operation, serial data are transferred in the unit of 9 bits. After CS# pull low, the first byte is command byte, the D/C# bit is as 0 and following with the register byte. After command byte send, the following byte(s) are data byte(s), with D/C# bit is 1. After D/C# bit sending from MCU, an 8-bit data will be shifted out on every clock falling edge. The serial data output SDO bit shifting sequence is D7, D6, to D0 bit. Figure 7-4 shows the read procedure in 3-wire SPI.

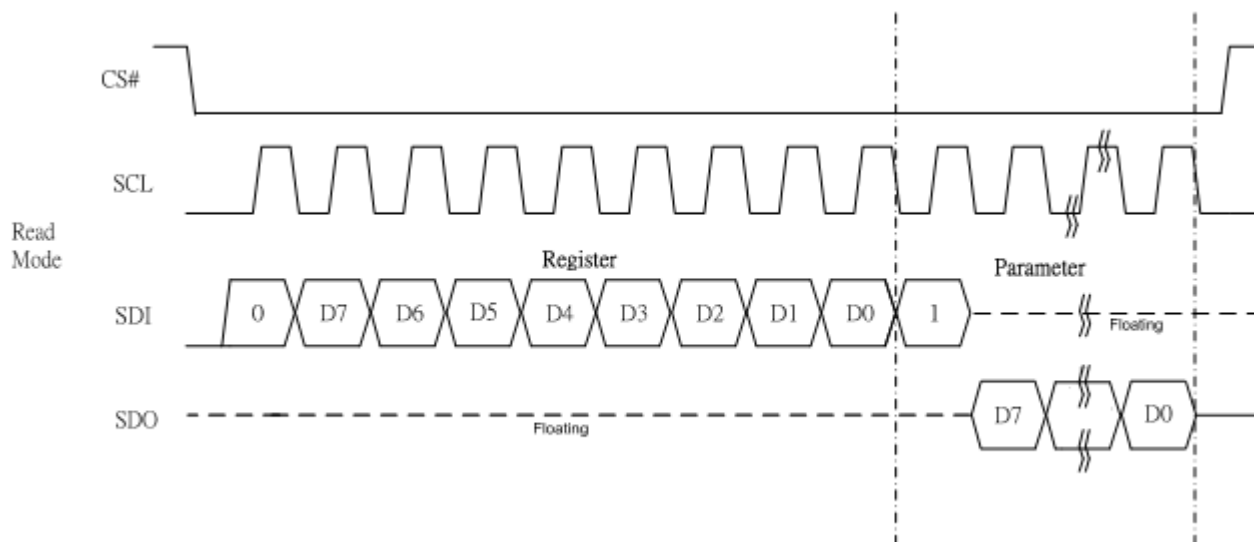


Figure 7-4: Read procedure in 3-wire SPI mode



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8. COMMAND TABLE

Command Table																																																								
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																												
0	0	01	0	0	0	0	0	0	0	1	Driver Output control	Gate setting A[9:0]= 2A7h [POR], 680 MUX MUX Gate lines setting as (A[9:0] + 1).																																												
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																														
0	1		0	0	0	0	0	0	A9	A8																																														
0	1		0		0	0	0	B2	B1	B0																																														
												B[2]: GD Selects the 1st output Gate GD=0 [POR], G0 is the 1st gate output channel, gate output sequence is G0,G1, G2, G3, ... GD=1, G1 is the 1st gate output channel, gate output sequence is G1, G0, G3, G2, ... B[1]: SM Change scanning order of gate driver. SM=0 [POR], G0, G1, G2, G3...679 (left and right gate interlaced) SM=1, G0, G2, G4 ...G678, G1, G3, ...G679 B[0]: TB TB = 0 [POR], scan from G0 to G679 TB = 1, scan from G679 to G0.																																												
0	0	03	0	0	0	0	0	0	1	1	Gate Driving voltage Control	Set Gate driving voltage A[4:0] = 00h [POR] VGH setting from 12V to 20V																																												
												<table><tr><th>A[4:0]</th><th>VGH</th><th>A[4:0]</th><th>VGH</th></tr><tr><td>00h</td><td>20</td><td>10h</td><td>16.5</td></tr><tr><td>07h</td><td>12</td><td>11h</td><td>17</td></tr><tr><td>08h</td><td>12.5</td><td>12h</td><td>17.5</td></tr><tr><td>09h</td><td>13</td><td>13h</td><td>18</td></tr><tr><td>0Ah</td><td>13.5</td><td>14h</td><td>18.5</td></tr><tr><td>0Bh</td><td>14</td><td>15h</td><td>19</td></tr><tr><td>0Ch</td><td>14.5</td><td>16h</td><td>19.5</td></tr><tr><td>0Dh</td><td>15</td><td>17h</td><td>20</td></tr><tr><td>0Eh</td><td>15.5</td><td>Other</td><td>NA</td></tr><tr><td>0Fh</td><td>16</td><td></td><td></td></tr></table>	A[4:0]	VGH	A[4:0]	VGH	00h	20	10h	16.5	07h	12	11h	17	08h	12.5	12h	17.5	09h	13	13h	18	0Ah	13.5	14h	18.5	0Bh	14	15h	19	0Ch	14.5	16h	19.5	0Dh	15	17h	20	0Eh	15.5	Other	NA	0Fh	16		
A[4:0]	VGH	A[4:0]	VGH																																																					
00h	20	10h	16.5																																																					
07h	12	11h	17																																																					
08h	12.5	12h	17.5																																																					
09h	13	13h	18																																																					
0Ah	13.5	14h	18.5																																																					
0Bh	14	15h	19																																																					
0Ch	14.5	16h	19.5																																																					
0Dh	15	17h	20																																																					
0Eh	15.5	Other	NA																																																					
0Fh	16																																																							



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Command Table														
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description		
0	0	04	0	0	0	0	0	1	0	0	Source Driving voltage Control	Set Source driving voltage A[7:0] = 41h [POR], VSH1 at 15V B[7:0] = A8h [POR], VSH2 at 5V. C[7:0] = 32h [POR], VSL at -15V		
0	1		A7	A6	A5	A4	A3	A2	A1	A2				
0	1		B7	B6	B5	B4	B3	B2	B1	B0				
0	1		C7	C6	C5	C4	C3	C2	C1	C0				
B[7] = 1, VSH2 voltage setting from 2.4V to 8.8V			A[7]/B[7] = 0, VSH1/VSH2 voltage setting from 9V to 17V					C[7] = 0, VSL setting from -9V to -17V						
A/B[7:0]			VSH1/VSH2			A/B[7:0]			VSH1/VSH2			C[7:0]		VSL
8Eh			2.4			AFh			5.7			1Ah		-9
8Fh			2.5			B0h			5.8			1Ch		-9.5
90h			2.6			B1h			5.9			1Eh		-10
91h			2.7			B2h			6			20h		-10.5
92h			2.8			B3h			6.1			22h		-11
93h			2.9			B4h			6.2			24h		-11.5
94h			3			B5h			6.3			26h		-12
95h			3.1			B6h			6.4			28h		-12.5
96h			3.2			B7h			6.5			2Ah		-13
97h			3.3			B8h			6.6			2Ch		-13.5
98h			3.4			B9h			6.7			2Eh		-14
99h			3.5			BAh			6.8			30h		-14.5
9Ah			3.6			BBh			6.9			32h		-15
9Bh			3.7			BCh			7			34h		-15.5
9Ch			3.8			BDh			7.1			36h		-16
9Dh			3.9			BEh			7.2			38h		-16.5
9Eh			4			BFh			7.3			3Ah		-17
9Fh			4.1			C0h			7.4			Other		NA
A0h			4.2			C1h			7.5					
A1h			4.3			C2h			7.6					
A2h			4.4			C3h			7.7					
A3h			4.5			C4h			7.8					
A4h			4.6			C5h			7.9					
A5h			4.7			C6h			8					
A6h			4.8			C7h			8.1					
A7h			4.9			C8h			8.2					
A8h			5			C9h			8.3					
A9h			5.1			CAh			8.4					
AAh			5.2			CBh			8.5					
ABh			5.3			CCh			8.6					
ACh			5.4			CDh			8.7					
ADh			5.5			CEh			8.8					
AEh			5.6			Other			NA					

Remark:
VSH1 > VSH2



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R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command		Description	
0	0	10	0	0	0	1	0	0	0	0	Deep Sleep mode	Sleep	Deep Sleep mode Control:	
0	0		0	0	0	0	0	A2	A1	A0			A[1:0] :	Description
													00	Normal Mode [POR]
													01	Enter Deep Sleep Mode 1
											11	Enter Deep Sleep Mode 2		
After this command initiated, the chip will enter Deep Sleep Mode, BUSY pad will keep output high. Remark: To Exit Deep Sleep mode, User required to send HWRESET to the driver														
0	0	11	0	0	0	1	0	0	0	1	Data mode setting	Entry	Define data entry sequence	
0	1		0	0	0	0	0	A2	A1	A0			A[2:0] = 011 [POR] A [1:0] = ID[1:0] Address automatic increment / decrement setting The setting of incrementing or decrementing of the address counter can be made independently in each upper and lower bit of the address. 00 –Y decrement, X decrement, 01 –Y decrement, X increment, 10 –Y increment, X decrement, 11 –Y increment, X increment [POR] A[2] = AM Set the direction in which the address counter is updated automatically after data are written to the RAM. AM= 0, the address counter is updated in the X direction. [POR] AM = 1, the address counter is updated in the Y direction.	
0	0	12	0	0	0	1	0	0	1	0	SW RESET		It resets the commands and parameters to their S/W Reset default values except R10h-Deep Sleep Mode During operation, BUSY pad will output high. Note: RAM are unaffected by this command.	



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Command Table																										
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description														
0	0	14	0	0	0	1	0	1	0	0	HV Ready Detection	HV ready detection A[6:0] = 00h [POR] The command required CLKEN=1 and ANALOGEN=1. Refer to Register 0x22 for detail. After this command initiated, HV Ready detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).														
0	0	15	0	0	0	1	0	1	0	1	VCI Detection	VCI Detection A[2:0] = 100 [POR] , Detect level at 2.3V A[2:0] : VCI level Detect <table><tr><td>A[2:0]</td><td>VCI level</td></tr><tr><td>011</td><td>2.2V</td></tr><tr><td>100</td><td>2.3V</td></tr><tr><td>101</td><td>2.4V</td></tr><tr><td>110</td><td>2.5V</td></tr><tr><td>111</td><td>2.6V</td></tr><tr><td>Other</td><td>NA</td></tr></table> The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. After this command initiated, VCI detection starts. BUSY pad will output high during detection. The detection result can be read from the Status Bit Read (Command 0x2F).	A[2:0]	VCI level	011	2.2V	100	2.3V	101	2.4V	110	2.5V	111	2.6V	Other	NA
A[2:0]	VCI level																									
011	2.2V																									
100	2.3V																									
101	2.4V																									
110	2.5V																									
111	2.6V																									
Other	NA																									
0	1		0	0	0	0	0	A2	A1	A0																
0	0	18	0	0	0	1	1	0	0	0	Temperature Sensor Control	Temperature Sensor Selection A[7:0] = 48h [POR], external temperature sensor A[7:0] = 80h Internal temperature sensor														
0	1		A7	A6	A5	A4	A3	A2	A1	A0																
0	0	1A	0	0	0	0	0	0	1	0	Temperature Sensor Control (Write to temperature register)	Write to temperature register. A[11:0] = 7FFh[POR]														
0	1		A11	A10	A9	A8	A7	A6	A5	A4																
0	1		A3	A2	A1	A0	0	0	0	0																
0	0	1B	0	0	0	1	1	0	1	1	Temperature Sensor Control (Read from temperature register)	Read from temperature register.														
1	1		A11	A10	A9	A8	A7	A6	A5	A4																
1	1		A3	A2	A1	A0	0	0	0	0																



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Command Table																								
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description												
0	0	1C	0	0	0	1	1	1	0	0	Temperature Sensor Control (Write Command to External temperature sensor)	Write Command to External temperature sensor. A[7:0] = 00h [POR], B[7:0] = 00h [POR], C[7:0] = 00h [POR], A[7:6] <table><tr><td>A[7:6]</td><td>Select no of byte to be sent</td></tr><tr><td>00</td><td>Address + pointer</td></tr><tr><td>01</td><td>Address + pointer + 1st arameter</td></tr><tr><td>10</td><td>Address + pointer + 1st parameter +2nd pointer</td></tr><tr><td>11</td><td>Address</td></tr></table> A[5:0] – Pointer Setting B[7:0] – 1st parameter C[7:0] – 2nd parameter The command required CLKEN=1. Refer to Register 0x22 for detail. After this command initiated, Write Command to external temperature sensor starts. BUSY pad will output high during operation.	A[7:6]	Select no of byte to be sent	00	Address + pointer	01	Address + pointer + 1st arameter	10	Address + pointer + 1st parameter +2nd pointer	11	Address		
A[7:6]	Select no of byte to be sent																							
00	Address + pointer																							
01	Address + pointer + 1st arameter																							
10	Address + pointer + 1st parameter +2nd pointer																							
11	Address																							
0	1		A7	A6	A5	A4	A3	A2	A1	A0														
0	1		B7	B6	B5	B4	B3	B2	B1	B0														
0	1		C7	C6	C5	C4	C3	C2	C1	C0														
0	0	20	0	0	1	0	0	0	0	0	Master Activation	Activate Display Update Sequence The Display Update Sequence Option is located at R22h. BUSY pad will output high during operation. User should not interrupt this operation to avoid corruption of panel images.												
0	0	21	0	0	1	0	0	0	0	1	Display Update Control 1	RAM content option for Display Update A[7:0] = 00h [POR] A[7:4] Red RAM option <table><tr><td>0000</td><td>Normal</td></tr><tr><td>0100</td><td>Bypass RAM content as 0</td></tr><tr><td>1000</td><td>Inverse RAM content</td></tr></table> A[3:0] BW RAM option <table><tr><td>0000</td><td>Normal</td></tr><tr><td>0100</td><td>Bypass RAM content as 0</td></tr><tr><td>1000</td><td>Inverse RAM content</td></tr></table>	0000	Normal	0100	Bypass RAM content as 0	1000	Inverse RAM content	0000	Normal	0100	Bypass RAM content as 0	1000	Inverse RAM content
0000	Normal																							
0100	Bypass RAM content as 0																							
1000	Inverse RAM content																							
0000	Normal																							
0100	Bypass RAM content as 0																							
1000	Inverse RAM content																							
0	1		A7	A6	A5	A4	A3	A2	A1	A0														
0	0	22	0	0	1	0	0	0	1	0	Display Update Control 2	Display Update Sequence Option: Enable the stage for Master Activation A[7:0]= FFh (POR) <table><tr><td></td><td>Parameter (in Hex)</td></tr><tr><td>Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 1 Then Disable ANALOG Then Disable OSC</td><td>C7</td></tr><tr><td>Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 2 Then Disable ANALOG Then Disable OSC</td><td>CF</td></tr><tr><td colspan="2"></td></tr><tr><td>Enable Clock Signal, Then Load LUT with DISPLAY Mode 1</td><td>90</td></tr><tr><td>Enable Clock Signal, Then Load Temperature value from I2C Single Master Interface Then Load LUT with DISPLAY Mode 1</td><td>B0</td></tr></table>		Parameter (in Hex)	Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 1 Then Disable ANALOG Then Disable OSC	C7	Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 2 Then Disable ANALOG Then Disable OSC	CF			Enable Clock Signal, Then Load LUT with DISPLAY Mode 1	90	Enable Clock Signal, Then Load Temperature value from I2C Single Master Interface Then Load LUT with DISPLAY Mode 1	B0
	Parameter (in Hex)																							
Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 1 Then Disable ANALOG Then Disable OSC	C7																							
Enable Clock Signal, Then Enable ANALOG Then DISPLAY with DISPLAY Mode 2 Then Disable ANALOG Then Disable OSC	CF																							
Enable Clock Signal, Then Load LUT with DISPLAY Mode 1	90																							
Enable Clock Signal, Then Load Temperature value from I2C Single Master Interface Then Load LUT with DISPLAY Mode 1	B0																							
0	1		A7	A6	A5	A4	A3	A2	A1	A0														

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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D ₃	D2	D1	D0	Command	Description
0	0	24	0	0	1	0	0	1	0	0	Write RAM (BW)	After this command, data entries will be written into the BW RAM until another command is written. Address pointers will advance accordingly. For Write pixel: Content of Write RAM(BW) = 1 For Black pixel: Content of Write RAM(BW) = 0
0	0	26	0	0	1	0	0	1	1	0	Write RAM (RED)	After this command, data entries will be written into the RED RAM until another command is written. Address pointers will advance accordingly. For Red pixel: Content of Write RAM(RED) = 1 For non-Red pixel [Black or White]: Content of Write RAM(RED) = 0
0	0	27	0	0	1	0	0	1	1	1	Read RAM	After this command, data read on the MCU bus will fetch data from RAM [According to parameter of Register 41h to select reading RAM(BW) / RAM(RED)], until another command is written. Address pointers will advance accordingly. The 1st byte of data read is dummy data.
0	0	28	0	0	1	0	1	0	0	0	VCOM Sense	Enter VCOM sensing conditions and hold for duration defined in 29h before reading VCOM value. The sensed VCOM voltage is stored in register The command required CLKEN=1 and ANALOGEN=1 Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	29	0	0	1	0	1	0	0	1	VCOM Sense Duration	Stabling time between entering VCOM sensing mode and reading acquired. A[6]=1, Normal Mode A[6]=0, Reserve A[3:0] = 09h, duration = 10s. VCOM sense duration = Setting + 1 Seconds
0	1		0	A6	0	0	A3	A2	A1	A0		
0	0	2A	0	0	1	0	1	0	1	0	Program VCOM OTP	Program VCOM register into OTP The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.



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Command Table																																																																																																
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																																																																				
0	0	2B	0	0	1	0	1	0	1	1	Write Register for VCOM Control	This command is used to reduce glitch when ACVCOM toggle. Two data bytes D04h and D63h should be set for this command.																																																																																				
0	1		0	0	0	0	0	1	0	0																																																																																						
0	1		0	1	1	0	0	0	1	1																																																																																						
0	0	2C	0	0	1	0	1	1	0	0	Write VCOM register	Write VCOM register from MCU interface A[7:0] = 00h [POR]																																																																																				
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																																																																						
												<table><tr><th>A[7:0]</th><th>VCOM</th><th>A[7:0]</th><th>VCOM</th></tr><tr><td>08h</td><td>-0.2</td><td>58h</td><td>-2.2</td></tr><tr><td>0Ch</td><td>-0.3</td><td>5Ch</td><td>-2.3</td></tr><tr><td>10h</td><td>-0.4</td><td>60h</td><td>-2.4</td></tr><tr><td>14h</td><td>-0.5</td><td>64h</td><td>-2.5</td></tr><tr><td>18h</td><td>-0.6</td><td>68h</td><td>-2.6</td></tr><tr><td>1Ch</td><td>-0.7</td><td>6Ch</td><td>-2.7</td></tr><tr><td>20h</td><td>-0.8</td><td>70h</td><td>-2.8</td></tr><tr><td>24h</td><td>-0.9</td><td>74h</td><td>-2.9</td></tr><tr><td>28h</td><td>-1</td><td>78h</td><td>-3</td></tr><tr><td>2Ch</td><td>-1.1</td><td>7Ch</td><td>-3.1</td></tr><tr><td>30h</td><td>-1.2</td><td>80h</td><td>-3.2</td></tr><tr><td>34h</td><td>-1.3</td><td>84</td><td>-3.3</td></tr><tr><td>38h</td><td>-1.4</td><td>88</td><td>-3.4</td></tr><tr><td>3Ch</td><td>-1.5</td><td>8C</td><td>-3.5</td></tr><tr><td>40h</td><td>-1.6</td><td>90</td><td>-3.6</td></tr><tr><td>44h</td><td>-1.7</td><td>94</td><td>-3.7</td></tr><tr><td>48h</td><td>-1.8</td><td>98</td><td>-3.8</td></tr><tr><td>4Ch</td><td>-1.9</td><td>9C</td><td>-3.9</td></tr><tr><td>50h</td><td>-2</td><td>A0</td><td>-4</td></tr><tr><td>54h</td><td>-2.1</td><td></td><td></td></tr></table>	A[7:0]	VCOM	A[7:0]	VCOM	08h	-0.2	58h	-2.2	0Ch	-0.3	5Ch	-2.3	10h	-0.4	60h	-2.4	14h	-0.5	64h	-2.5	18h	-0.6	68h	-2.6	1Ch	-0.7	6Ch	-2.7	20h	-0.8	70h	-2.8	24h	-0.9	74h	-2.9	28h	-1	78h	-3	2Ch	-1.1	7Ch	-3.1	30h	-1.2	80h	-3.2	34h	-1.3	84	-3.3	38h	-1.4	88	-3.4	3Ch	-1.5	8C	-3.5	40h	-1.6	90	-3.6	44h	-1.7	94	-3.7	48h	-1.8	98	-3.8	4Ch	-1.9	9C	-3.9	50h	-2	A0	-4	54h	-2.1		
A[7:0]	VCOM	A[7:0]	VCOM																																																																																													
08h	-0.2	58h	-2.2																																																																																													
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10h	-0.4	60h	-2.4																																																																																													
14h	-0.5	64h	-2.5																																																																																													
18h	-0.6	68h	-2.6																																																																																													
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38h	-1.4	88	-3.4																																																																																													
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44h	-1.7	94	-3.7																																																																																													
48h	-1.8	98	-3.8																																																																																													
4Ch	-1.9	9C	-3.9																																																																																													
50h	-2	A0	-4																																																																																													
54h	-2.1																																																																																															
0	0	2D	0	0	1	0	1	1	0	1	OTP Register Read forDisplay Option	Read Register for Display Option: A[7:0]: VCOM OTP Selection (Command 0x37, Byte A) B[7:0]: VCOM Register (Command 0x2C) C[7:0]~G[7:0]: Display Mode (Command 0x37, Byte B to Byte G) [5 bytes] H[7:0]~K[7:0]: Waveform Version (Command 0x37, Byte H to Byte K) [4 bytes]																																																																																				
1	1		A7	A6	A5	A4	A3	A2	A1	A0																																																																																						
1	1		B7	B6	B5	B4	B3	B2	B1	B0																																																																																						
1	1		C7	C6	C5	C4	C3	C2	C1	C0																																																																																						
1	1		D7	D6	D5	D4	D3	D2	D1	D0																																																																																						
1	1		E7	E6	E5	E4	E3	E2	E1	E0																																																																																						
1	1		F7	F6	F5	F4	F3	F2	F1	F0																																																																																						
1	1		G7	G6	G5	G4	G3	G2	G1	G0																																																																																						
1	1		H7	H6	H5	H4	H3	H2	H1	H0																																																																																						
1	1		I7	I6	I5	I4	I3	I2	I1	I0																																																																																						
1	1		J7	J6	J5	J4	J3	J2	J1	J0																																																																																						
1	1		K7	K6	K5	K4	K3	K2	K1	K0																																																																																						



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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	34	0	0	1	1	0	1	0	0	CRC calculation	CRC calculation command for OTP content validation. BUSY pad will output high during operation.
0	0	35	0	0	1	1	0	1	0	1	CRC Status Read	CRC Status Read A[15:0] is the CRC readout Value
1	1		A15	A14	A13	A12	A11	A10	A9	A8		
1	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	0	36	0	0	1	1	0	1	1	0	Program selection OTP	Program OTP Selection according to the OTP Selection Control [R37h and R38h] The command required CLKEN=1. Refer to Register 0x22 for detail. BUSY pad will output high during operation.
0	0	37	0	0	1	1	0	1	1	1	Write Register for Display Option	Write Register for Display Option B[7:0] Display Mode for WS[7:0] C[7:0] Display Mode for WS[15:8] D[7:0] Display Mode for WS[23:16] E[7:0] Display Mode for WS[31:24] F[3:0] Display Mode for WS[35:32] 0: Display Mode 1 1: Display Mode2 F[6]: PingPong for Display Mode 2 F[7]: PingPong for Display Mode 1 1: Ping-Pong 0: Default G[7:0]~J[7:0] module ID /waveform version. Remarks: A[7:0]~J[7:0] can be stored in OTP
0	1		0	0	0	0	0	0	0	0		
0	1		B7	B6	B5	B4	B3	B2	B1	B0		
0	1		C7	C6	C5	C4	C3	C2	C1	C0		
0	1		D7	D6	D5	D4	D3	D2	D1	D0		
0	1		E7	E6	E5	E4	E3	E2	E1	E0		
0	1		F7	F6	F5	F4	F3	F2	F1	F0		
0	1		G7	G6	G5	G4	G3	G2	G1	G0		
0	1		H7	H6	H5	H4	H3	H2	H1	H0		
0	1		I7	I6	I5	I4	I3	I2	I1	I0		
0	1		J7	J6	J5	J4	J3	J2	J1	J0		
0	0	38	0	0	1	1	1	0	0	0	Write Register for User ID	Write Register for User ID A[7:0]~J[7:0]: UserID [10 bytes] Remarks: A[7:0]~J[7:0] can be stored in OTP
0	1	A7	A6	A5	A4	A3	A2	A1	A0	A7		
0	1	B7	B6	B5	B4	B3	B2	B1	B0	B7		
0	1	C7	C6	C5	C4	C3	C2	C1	C0	C7		
0	1	D7	D6	D5	D4	D3	D2	D1	D0	D7		
0	1	E7	E6	E5	E4	E3	E2	E1	E0	E7		
0	1	F7	F6	F5	F4	F3	F2	F1	F0	F7		
0	1	G7	G6	G5	G4	G3	G2	G1	G0	G7		
0	1	H7	H6	H5	H4	H3	H2	H1	H0	H7		
0	1	I7	I6	I5	I4	I3	I2	I1	I0	I7		
0	1	J7	J6	J5	J4	J3	J2	J1	J0	J7		



File Name			Specification For HINK 10.2 EPD								Module Number		HINK-E102A03	
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Command Table														
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description		
0	0	39	0	0	1	1	1	0	0	1	OTP program mode	OTP program mode		
0	1		0	0	0	0	0	0	A1	A0		A[1:0] = 00: Normal Mode [POR] A[1:0] = 11: Internal generated OTP programming voltage Remark: User is required to EXACTLY follow the reference code sequences		
0	0	3A	0	0	1	1	1	0	1	0	Reserved	Reserved		
0	0	3B	0	0	1	1	1	0	1	1	Reserved	Reserved		
0	0	3C	0	0	1	1	1	1	0	0	Border Waveform Control	Select border waveform for VBD A[7:0] = C0h [POR], set VBD as HiZ. A [7:6] :Select VBD option		
0	1		A7	A6	A5	A4	A3	A2	A1	A0		A[7:6] Select VBD as		
												00 GS Transition, Defined in A[1:0]		
												01 Fix Level, Defined in A[5:4]		
												10 VCOM		
												11[POR] HiZ		
												A [5:4] Fix Level Setting for VBD		
												A[5:4] VBD level		
												00[POR] VSS		
												01 VSH1		
												10 VSL		
												11 VSH2		
												A [1:0] GS Transition setting for VBD		
												A[1:0] VBD Transition		
												00[POR] LUT0		
											01 LUT1			
											10 LUT2			
											11 LUT3			
0	0	41	0	1	0	0	0	0	0	1	Read RAM Option	Read RAM Option		
0	1		0	0	0	0	0	0	0	A0		A[0]= 0 [POR] 0: Read RAM corresponding to 24h 1: Read RAM corresponding to 26h		
0	0	44	0	1	0	0	0	1	0	0	Set RAM X - address Start / End position	Specify the start/end positions of the window address in the X direction by an address unit for RAM		
0	1		A7	A6	A5	A4	A3	A2	A1	A0		A[9:0]: XSA[9:0], XStart, POR = 000h		
0	1		-	-	-	-	-	-	-	A9		A8	B[5:0]: XEA[9:0], XEnd, POR = 3BFh	
0	1		0	0	B5	B4	B3	B2	B1	B0				
0	1		-	-	-	-	-	-	-	B9		B8		



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Command Table																																																				
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description																																								
0	0	45	0	1	0	0	0	1	0	1	Set RAM Y-address Start / End position	Specify the start/end positions of the window address in the Y direction by an address unit for RAM A[8:0]: YSA[8:0], YStart, POR = 000h B[8:0]: YEA[8:0], YEnd, POR = 2A7h																																								
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																										
0	1		-	-	-	-	-	-	A9	A8																																										
0	1		B7	B6	B5	B4	B3	B2	B1	B0																																										
0	1		-	-	-	-	-	-	B9	B8																																										
0	0	46	0	1	0	0	0	1	1	0	Auto Write RED RAM for Regular Pattern	Auto Write RED RAM for Regular Pattern A[7:0] = 00h [POR] A[7]: The 1st step value, POR = 0 A[6:4]: Step Height, POR= 000 Step of alter RAM in Y-direction according to Gate <table><tr><th>A[6:4]</th><th>Height</th><th>A[6:4]</th><th>Height</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><th>A[2:0]</th><th>Width</th><th>A[2:0]</th><th>Width</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> BUSY pad will output high during operation.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680
A[6:4]	Height	A[6:4]	Height																																																	
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A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
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011	64	111	680																																																	
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																										
0	0	47	0	1	0	0	0	1	1	1	Auto Write B/W RAM for Regular Pattern	Auto Write B/W RAM for Regular Pattern A[7:0] = 00h [POR] Auto Write B/W RAM for Regular Pattern A[7:0] = 00h [POR] <table><tr><th>A[6:4]</th><th>Height</th><th>A[6:4]</th><th>Height</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>960</td></tr></table> A[2:0]: Step Width, POR= 000 Step of alter RAM in X-direction according to Source <table><tr><th>A[2:0]</th><th>Width</th><th>A[2:0]</th><th>Width</th></tr><tr><td>000</td><td>8</td><td>100</td><td>128</td></tr><tr><td>001</td><td>16</td><td>101</td><td>256</td></tr><tr><td>010</td><td>32</td><td>110</td><td>512</td></tr><tr><td>011</td><td>64</td><td>111</td><td>680</td></tr></table> uring operation, BUSY pad will output high.	A[6:4]	Height	A[6:4]	Height	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	960	A[2:0]	Width	A[2:0]	Width	000	8	100	128	001	16	101	256	010	32	110	512	011	64	111	680
A[6:4]	Height	A[6:4]	Height																																																	
000	8	100	128																																																	
001	16	101	256																																																	
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A[2:0]	Width	A[2:0]	Width																																																	
000	8	100	128																																																	
001	16	101	256																																																	
010	32	110	512																																																	
011	64	111	680																																																	
0	1		A7	A6	A5	A4	A3	A2	A1	A0																																										



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Command Table												
R/W#	D/C#	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0	0	4E	0	1	0	0	1	1	1	0	Set RAM X address counter	Make initial settings for the RAM X address in the address counter (AC) A[9:0]: 000h [POR].
0	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	1		0	0	0	0	0	0	A9	A8		
0	0	4F	0	1	0	0	1	1	1	1	Set RAM Y address counter	Make initial settings for the RAM Y address in the address counter (AC) A[9:0]: 000h [POR].
0	1		A7	A6	A5	A4	A3	A2	A1	A0		
0	1		0	0	0	0	0	0	A9	A8		
0	0	7F	0	1	1	1	1	1	1	1	NOP	This command is an empty command; it does not have any effect on the display module. However, it can be used to terminate Frame Memory Write or Read Commands.



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9.Reference Circuit

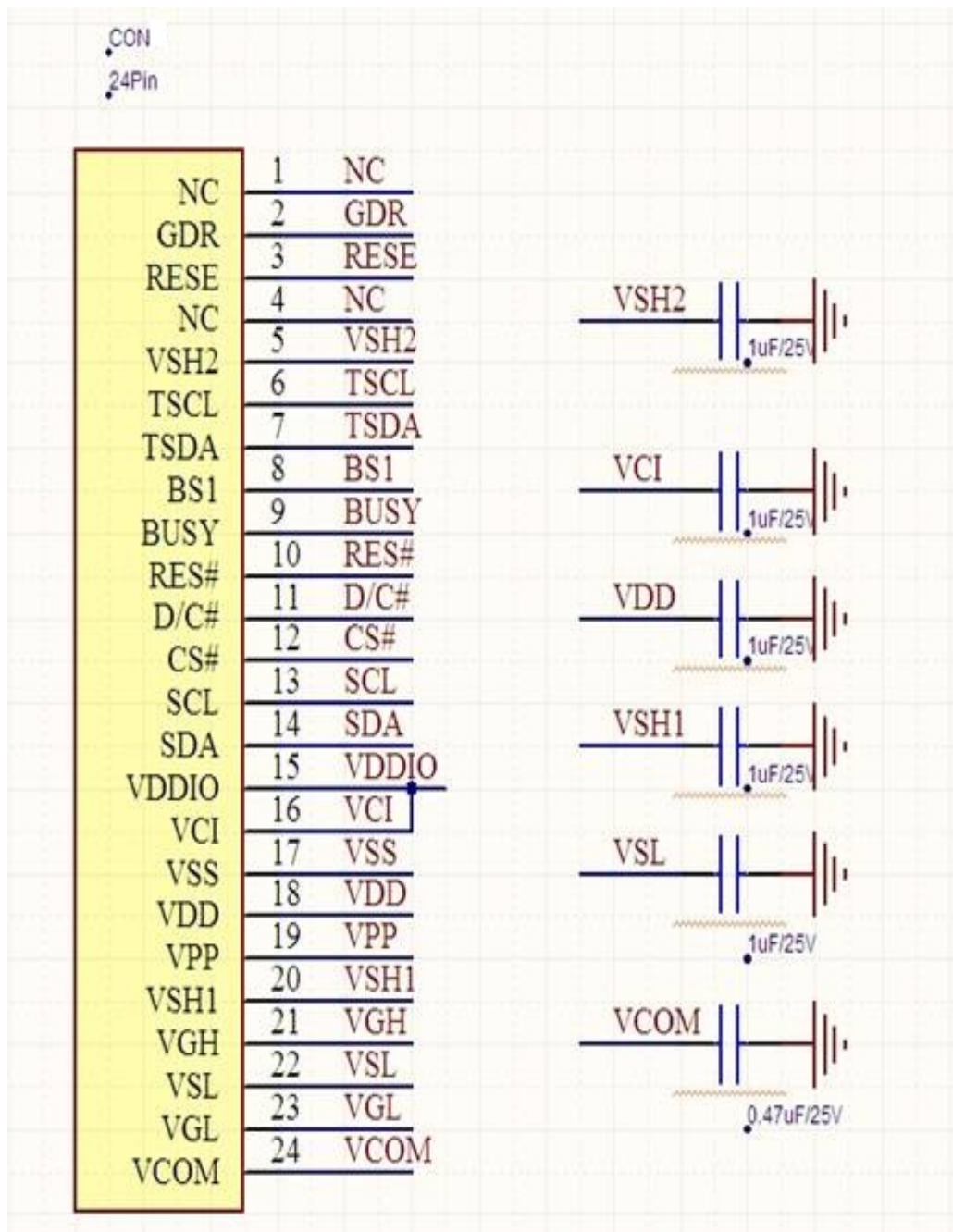


Figure. 9-1



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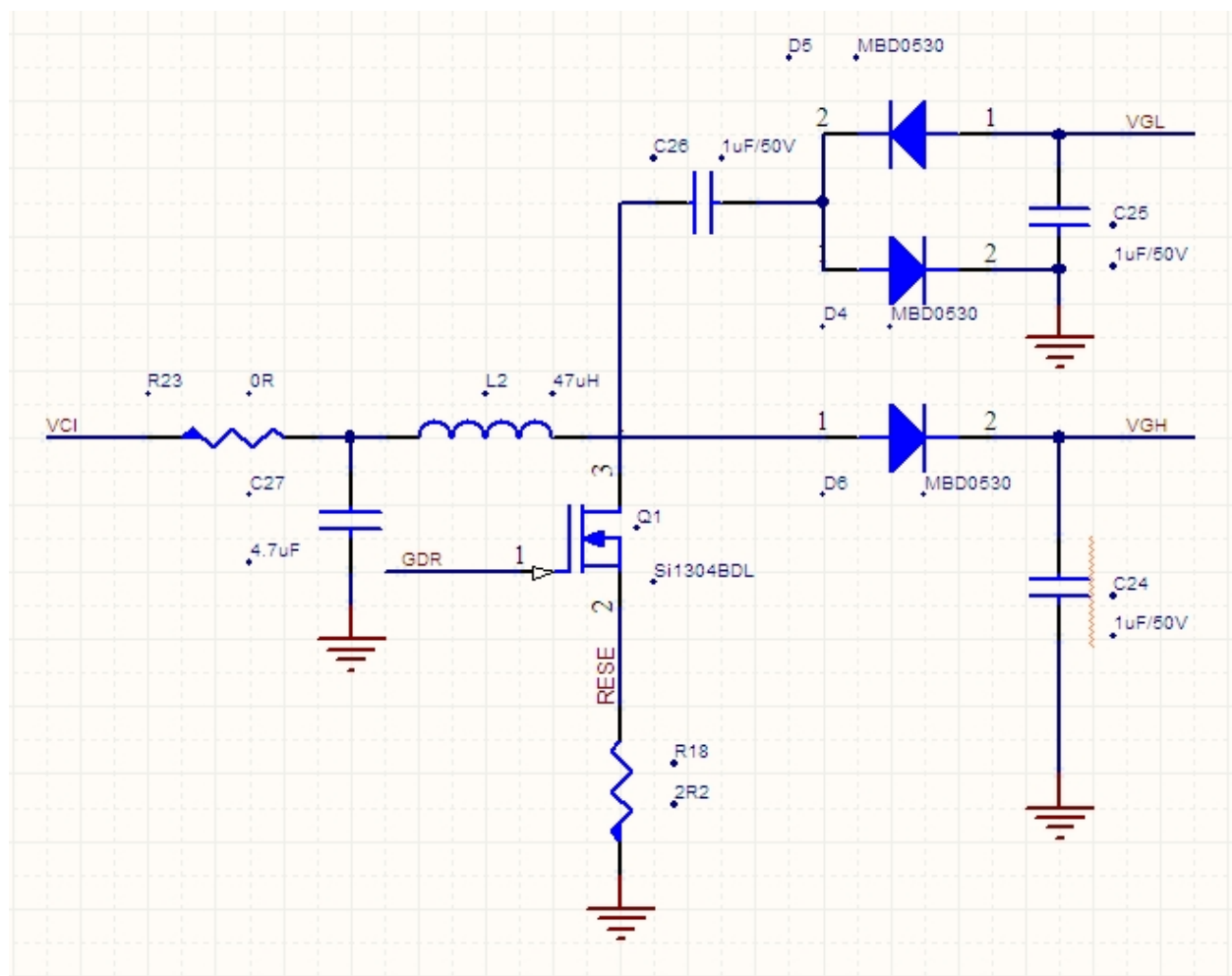


Figure. 9-2

Part Name	Value/Type	Reference Part
D4-D6	Diode	OnSemi:MBR0530
L2	47uH	Sumida:CDRH2D18/LDNP-470NC
Q1	NMOS	Vishay:Si1304BDL
R18	2.2Ohm	Vishay:CRCW08052R20FKEA
CON 24Pin	0.5mm ZIF socket	Hirose:FH34S-24S-0.5SH(50)



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10. ABSOLUTE MAXIMUM RATING

Table 11-1: Maximum Ratings

Symbol	Parameter	Rating	Unit	Humidity	Unit	Note
V _{CI}	Logic supply voltage	-0.5 to +6.0	V	-	-	
T _{OPR}	Operation temperature range	0 to 40	°C	35 to70	%	
T _{tgt}	Transportation temperature range	-25 to 60	°C	35 to70	-	Note10-2
T _{stg}	Storage condition	0 to 40	°C	35 to70	%	Maximum storage time: 5 years
-	After opening the package	0 to 40	°C	35 to70	%	

Note 10-1:Maximum ratings are those values beyond which damages to the device may occur.

Functional operation should be restricted to the limits in the Electrical Characteristics chapter.

Note10-2: T_{tgt} is the transportation condition, the transport time is within 10 days for -25°C~0°C or 50°C~60°C.

11. DC CHARACTERISTICS

The following specifications apply for: V_{SS}=0V, V_{CI}=3.3V, T_{OPR}=25°C.

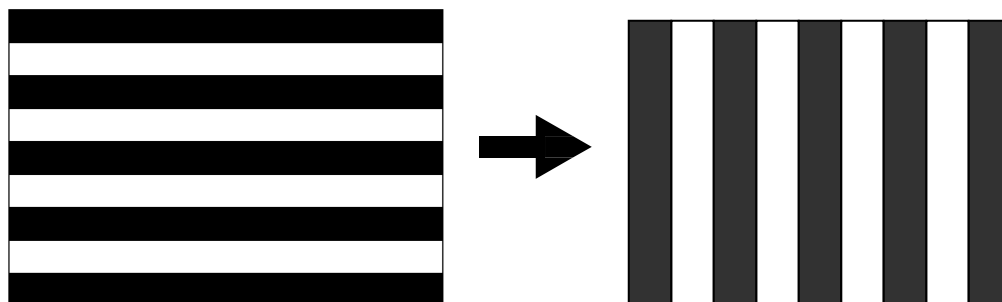
Table 12-1: DC Characteristics

Symbol	Parameter	Test Condition	Applicable pin	Min.	Typ.	Max.	Unit
V _{CI}	V _{CI} operation voltage	-	V _{CI}	2.2	3.3	3.7	V
V _{IH}	High level input voltage	-	SDA, SCL, CS#, D/C#,	0.8V _{DDIO}	-	-	V
V _{IL}	Low level input voltage	-	RES#, BS1	-	-	0.2V _{DDIO}	V
V _{OH}	High level output voltage	I _{OH} = -100uA	BUSY,	0.9V _{DDIO}	-	-	V
V _{OL}	Low level output voltage	I _{OL} = 100uA		-	-	0.1V _{DDIO}	V
I _{update}	Module operating current	-	-	-	8	-	mA
I _{sleep}	Deep sleep mode	V _{CI} =3.3V	-	-	-	3	uA

- The Typical power consumption is measured using associated 25°C waveform with following pattern transition: from horizontal scan pattern to vertical scan pattern. (Note 11-1)
- The listed electrical/optical characteristics are only guaranteed under the controller & waveform provided by XingTai.
- V_{com} value will be OTP before in factory or present on the label sticker.

Note 11-1

The Typical power consumption





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12. Serial Peripheral Interface Timing

The following specifications apply for: VSS=0V, VCI=2.2V to 3.7V, T_{OPR}=25°C

Write mode

Symbol	Parameter	Min	Typ	Max	Unit
fSCL	SCL frequency (Write Mode)			20	MHz
tCSSU	Time CSB has to be low before the first rising edge of SCLK	20			ns
tCSHLD	Time CSB has to remain low after the last falling edge of SCLK	20			ns
tCSHIGH	Time CSB has to remain high between two transfers	100			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	25			ns
tSCLLOW	Part of the clock period where SCL has to remain low	25			ns
tSISU	Time SI (SDA Write Mode) has to be stable before the next rising edge of SCL	10			ns
tSIHLD	Time SI (SDA Write Mode) has to remain stable after the rising edge of SCL	40			ns

Read mode

Symbol	Parameter	Min	Typ	Max	Unit
fSCL	SCL frequency (Read Mode)			2.5	MHz
tCSSU	Time CSB has to be low before the first rising edge of SCLK	100			ns
tCSHLD	Time CSB has to remain low after the last falling edge of SCLK	50			ns
tCSHIGH	Time CSB has to remain high between two transfers	250			ns
tSCLHIGH	Part of the clock period where SCL has to remain high	180			ns
tSCLLOW	Part of the clock period where SCL has to remain low	180			ns
tSOSU	Time SO(SDA Read Mode) will be stable before the next rising edge of SCL		50		ns
tSOHLD	Time SO (SDA Read Mode) will remain stable after the falling edge of SCL		0		ns

Note: All timings are based on 20% to 80% of VDDIO-VSS

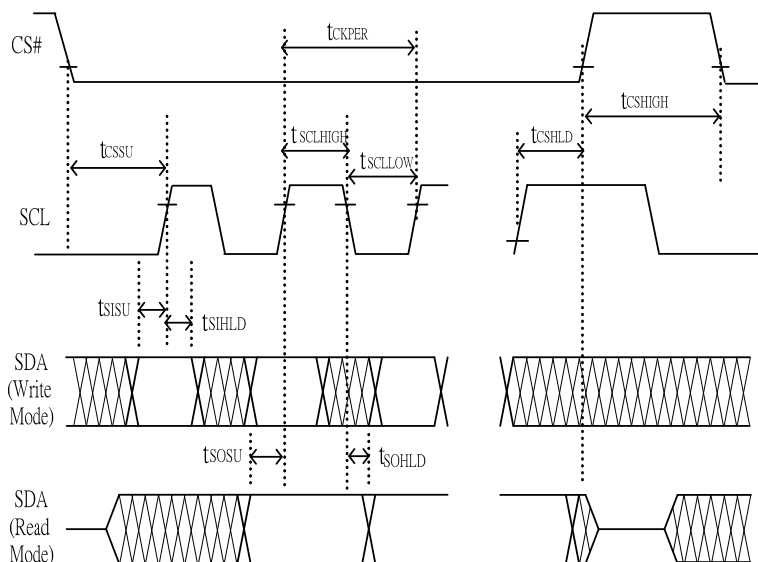


Figure 13-1: Serial peripheral interface characteristics

13. Power Consumption

Parameter	Symbol	Conditions	TYP	Max	Unit	Remark
Panel power consumption during update	-	25°C	-	35	mAs	-
Deep sleep mode	-	25°C	-	2	uA	-

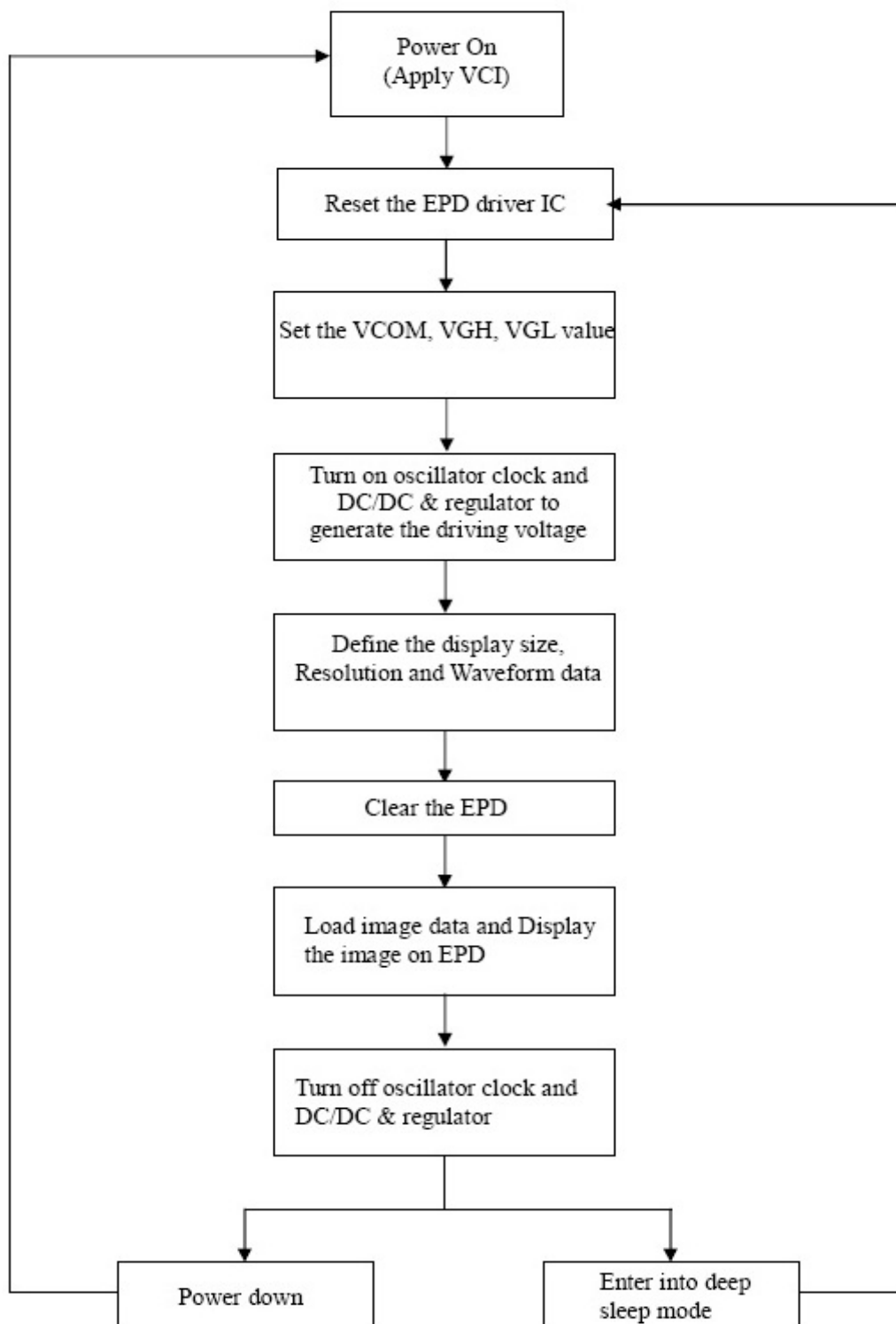
Mas=update average current × update time



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14. Typical Operating Sequence

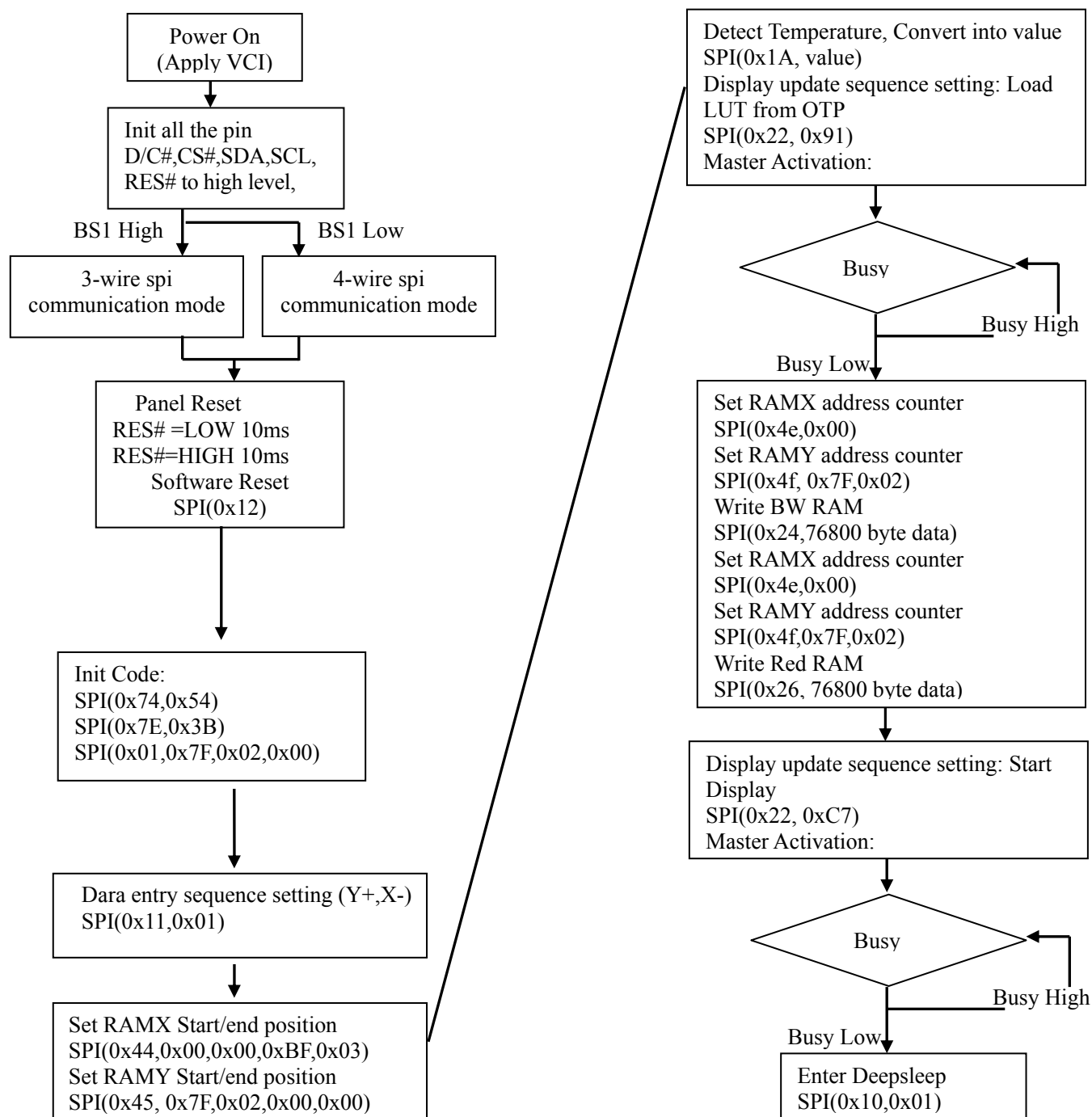
14.1 Normal Operation Flow





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14.2 Reference Program Code





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15. Optical characteristics

15.1 Specifications

Measurements are made with that the illumination is under an angle of 45 degrees, the detection is perpendicular unless otherwise specified.

T=25℃

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP.	MAX	UNIT	Note
R	Reflectance	White	30	35	-	%	Note 15-1
Gn	2Grey Level	-	-	$DS+(WS-DS) \times n(m-1)$	-	L*	-
CR	Contrast Ratio	-	-	10	-	-	-
KS	Black State L* value	-	-	18	-	-	Note 15-1
	Black State a* value	-	-	0.2	-	-	Note 15-1
WS	White State L* value	-	-	67	-	-	Note 15-1
Panel's life	-	0℃~30℃	-	5years	-	-	Note 15-2
Panel	Image Update	Storage and transportation	-	Update the white screen	-	-	-
	Update Time	Operation	-	Suggest Updated once a day	-	-	-

WS : White state, DS : Dark state

Note 15-1 : Luminance meter : i - One Pro Spectrophotometer

Note 15-2: We don't guarantee 5 years pixels display quality for humidity below 45%RH or above 70%RH;

at least update 1 time per day.



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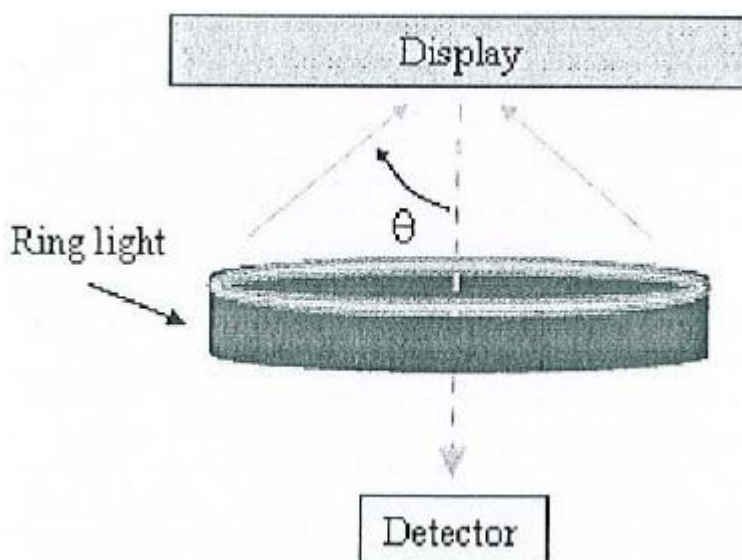
15.2 Definition of contrast ratio

The contrast ratio (CR) is the ratio between the reflectance in a full white area (R1) and the reflectance in a dark area (Rd):

R1: white reflectance

Rd: dark reflectance

$$CR = R1/Rd$$

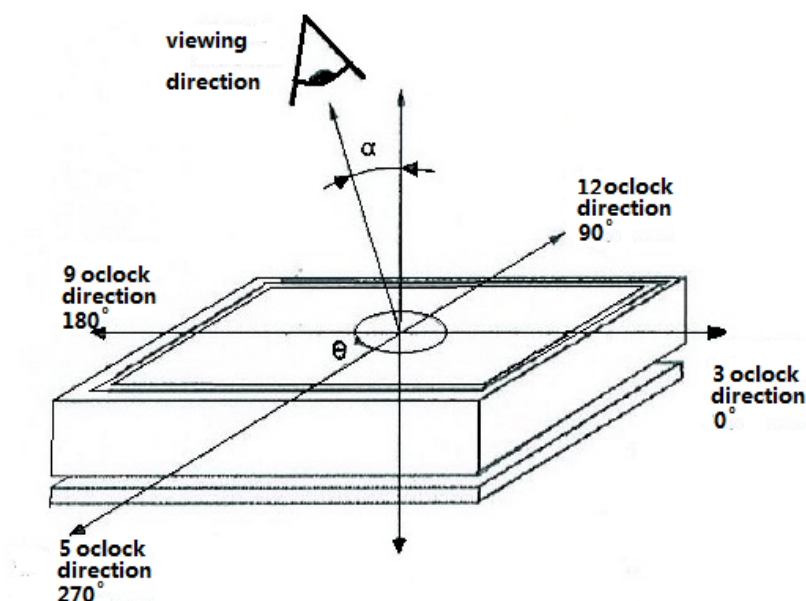


15.3 Reflection Ratio

The reflection ratio is expressed as :

$$R = \text{Reflectance Factor}_{\text{white board}} \times (L_{\text{center}} / L_{\text{white board}})$$

L_{center} is the luminance measured at center in a white area ($R=G=B=1$). $L_{\text{white board}}$ is the luminance of a standard white board. Both are measured with equivalent illumination source. The viewing angle shall be no more than 2 degrees.





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16. HANDLING,SAFETY AND ENVIROMENTAL REQUIREMENTS

WARNING
The display module should be kept flat or fixed to a rigid, curved support with limited bending along the long axis. It should not be used for continual flexing and bending. Handle with care. Should the display break do not touch any material that leaks out. In case of contact with the leaked material then wash with water and soap.

CAUTION
The display module should not be exposed to harmful gases, such as acid and alkali gases, which corrode electronic components.
Disassembling the display module can cause permanent damage and invalidate the warranty agreements.
IPA solvent can only be applied on active area and the back of a glass. For the rest part, it is not allowed.
Observe general precautions that are common to handling delicate electronic components. The glass can break and front surfaces can easily be damaged . Moreover the display is sensitive to static electricity and other rough environmental conditions.

Mounting Precautions
(1) It's recommended that you consider the mounting structure so that uneven force (ex. Twisted stress) is not applied to the module.
(2) It's recommended that you attach a transparent protective plate to the surface in order to protect the EPD. Transparent protective plate should have sufficient strength in order to resist external force.
(3) You should adopt radiation structure to satisfy the temperature specification.
(4) Acetic acid type and chlorine type materials for the cover case are not desirable because the former generates corrosive gas of attacking the PS at high temperature and the latter causes circuit break by electro-chemical reaction.
(5) Do not touch, push or rub the exposed PS with glass, tweezers or anything harder than HB pencil lead. And please do not rub with dust clothes with chemical treatment. Do not touch the surface of PS for bare hand or greasy cloth. (Some cosmetics deteriorate the PS)
(6) When the surface becomes dusty, please wipe gently with absorbent cotton or other soft materials like chamois soaks with petroleum benzene. Normal-hexane is recommended for cleaning the adhesives used to attach the PS. Do not use acetone, toluene and alcohol because they cause chemical damage to the PS.
(7) Wipe off saliva or water drops as soon as possible. Their long time contact with PS causes deformations and color fading.

Data sheet status	
Product specification	The data sheet contains final product specifications.



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Limiting values

Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.

Application information

Where application information is given, it is advisory and dose not form part of the specification.

Product Environmental certification

ROHS

REMARK

All The specifications listed in this document are guaranteed for module only. Post-assembled operation or component(s) may impact module performance or cause unexpected effect or damage and therefore listed specifications is not warranted after any Post-assembled operation.



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17. Reliability test

	TEST	CONDITION	REMARK
1	High-Temperature Operation	T=40℃, RH=35%RH, For 240Hr	
2	Low-Temperature Operation	T = 0℃ for 240 hrs	
3	High-Temperature Storage	T=60℃ RH=35%RH For 240Hr	Test in white pattern
4	Low-Temperature Storage	T = -25℃ for 240 hrs	Test in white pattern
5	High Temperature, High-Humidity Operation	T=40℃, RH=90%RH, For 168Hr	
6	High Temperature, High-Humidity Storage	T=60℃, RH=80%RH, For 240Hr Test in white pattern	Test in white pattern
7	Temperature Cycle	-25℃(30min)~70℃(30min), 100 Cycle	Test in white pattern
8	Package Vibration	1.04G,Frequency : 20~200Hz Direction : X,Y,Z Duration: 30 minutes in each direction	Full packed for shipment
9	Package Drop Impact	Drop from height of 100 cm on Concrete surface Drop sequence:1 corner, 3edges, 6face One drop for each.	Full packed for shipment
10	UV exposure Resistance	765 W/m ² for 168hrs,40℃	
11	Electrostatic discharge	Machine model: +/-250V,0Ω,200pF	

Actual EMC level to be measured on customer application.

Note1: The protective film must be removed before temperature test.

Note2: Stay white pattern for storage and non-operation test.

Note3: Operation is black/white/red pattern , hold time is 150S.

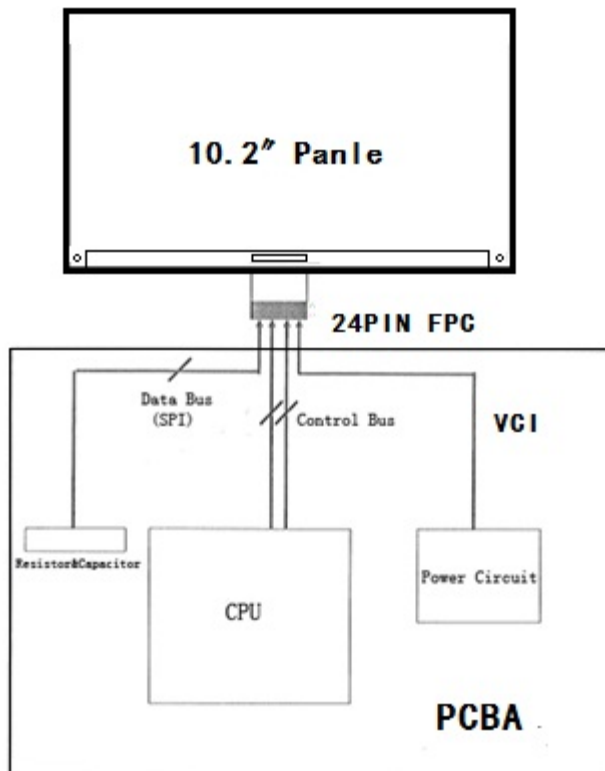
Note4: The function, appearance,opticals should meet the requirements of the test before and after the test.

Note5: Keep testing after 2 hours placing at 20℃-25℃.

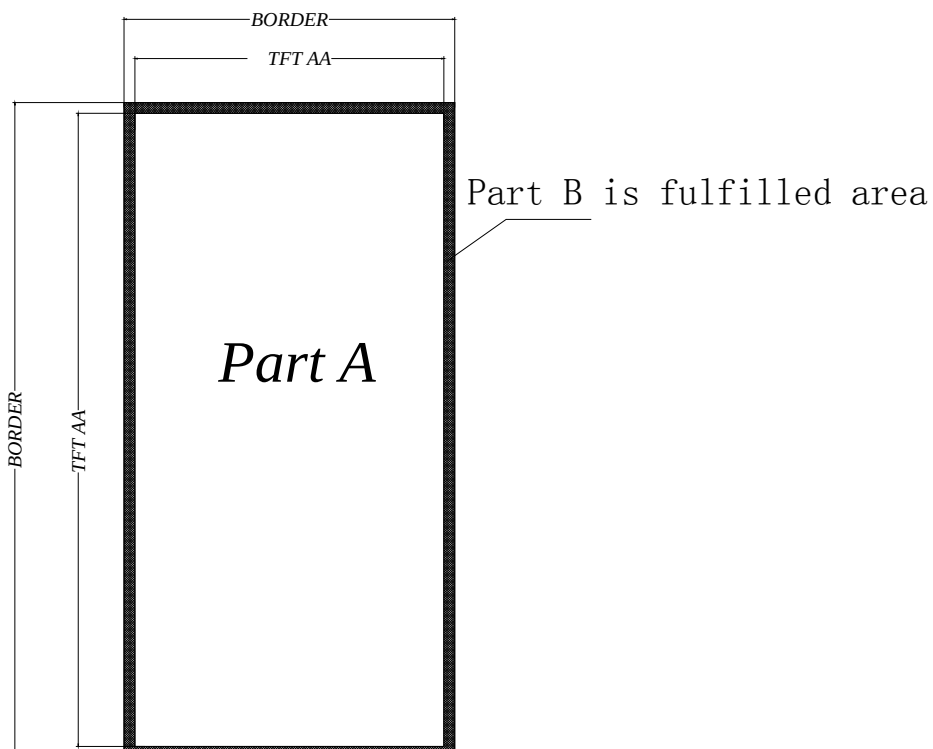


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18. Block Diagram



19. PartA/PartB specification



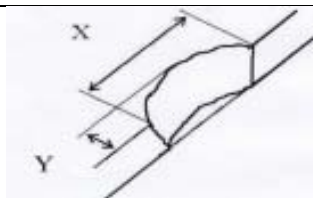


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20. Point and line standard

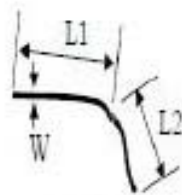
Shipment Inspection Standard

Equipment: Electrical test fixture, Point gauge

Outline dimension	224(H) × 157(V) × 1.1(D)	Unit: mm	Part-A	Active area	Part-B	Border area
Environment	Temperature	Humidity	Illuminance	Distance	Time	Angle
	19℃～25℃	55%±5%RH	800～1300Lux	300 mm	35Sec	
Defect type	Inspection method	Standard		Part-A		Part-B
Spot	Electric Display	D≤0.4 mm		Ignore		Ignore
		0.4 mm<D≤0.6 mm		N≤4		Ignore
		0.6 mm<D≤0.8 mm		N≤1		Ignore
		D>0.8 mm		Not Allow		Ignore
Display unwork	Electric Display	Not Allow		Not Allow		Ignore
Display error	Electric Display	Not Allow		Not Allow		Ignore
Scratch or line defect(include dirt)	Visual/Film card	L≤2 mm, W≤0.2 mm		Ignore		Ignore
		2.0mm<L≤9.0mm, 0.2<W≤0.6mm,		N≤2		Ignore
		L>9.0 mm, W>0.6 mm		Not Allow		Ignore
PS Bubble	Visual/Film card	D≤0.4mm		Ignore		Ignore
		0.4mm≤D≤0.6mm		N≤4		Ignore
		D>0.6 mm		Not Allow		Ignore
Side Fragment	Visual/Film card	X≤8mm, Y≤1mm, Do not affect the electrode circuit, Ignore				
						
Remark	1.Cannot be defect & failure cause by appearance defect;					
	2.Cannot be larger size cause by appearance defect;					
	L=long W=wide D=point size N=Defects NO					

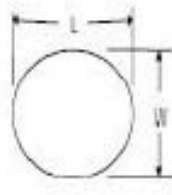


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$$L = L1 + L2$$

Line Defect



$$D = (L + W) / 2$$

Spot Defect

L=long W=wide D=point size



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21.Barcode

21-1 label appearance



ABBBBBBBCC
DDDEEEFGGG

21-2 QR scanned information (Total 28 code number+ 2 blank spaces)

A BBBBBBBB CC □ DDD EEE F GGG □ H III J KKK
① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨ ⑩ ⑪

- ① A——The factory code
- ② BBBBBBBB——Module name of EPD
- ③ CC——FPL model name
- ④ DDD——Date of production
- ⑤ EEE——Production lot
- ⑥ F——Separator
- ⑦ GGG——FPL Lot
- ⑧ H——Normal Lot
- ⑨ III——TFT、PS、EC.
- ⑩ J——IC
- ⑪ KKK——Serial NO.
- blank spaces



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22. Packing

TBD